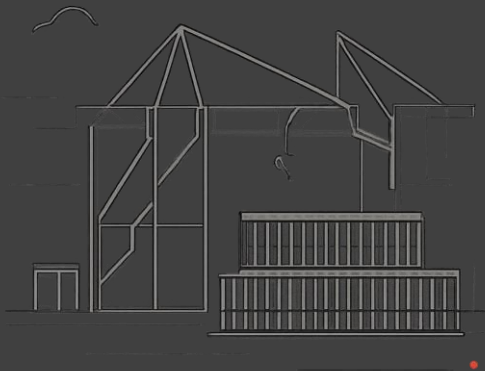




Supply Chain Shipment Pricing And Delivery Risk Analysis

An end-to-end analytics project analyzing pharmaceutical and health-commodity shipment data — modeling real-world global health supply chain challenges. Built to demonstrate SQL, Python, and Power BI skills on one realistic business problem.

GLOBAL HEALTH SUPPLY CHAIN

SCMS DATASET

SQL · PYTHON · POWER BI · AI

The Business Problem

A health-commodity supply chain organization ships pharmaceuticals and test kits to dozens of countries through multiple vendors and shipment modes. Ops and procurement leadership need clear answers to three critical questions:

Are We On Time?

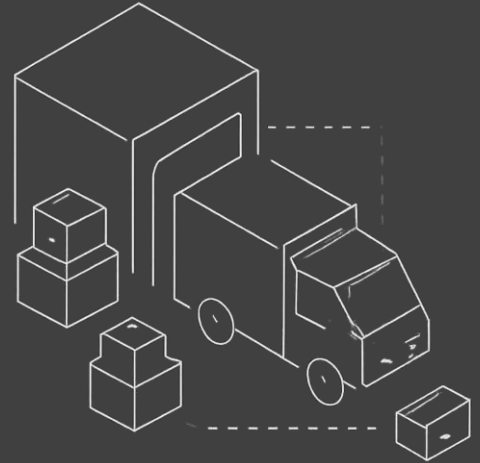
Are shipments arriving by their scheduled delivery dates — and where are we failing most?

What's Costing Us?

Which shipment modes and vendors are cost-efficient versus wasteful?

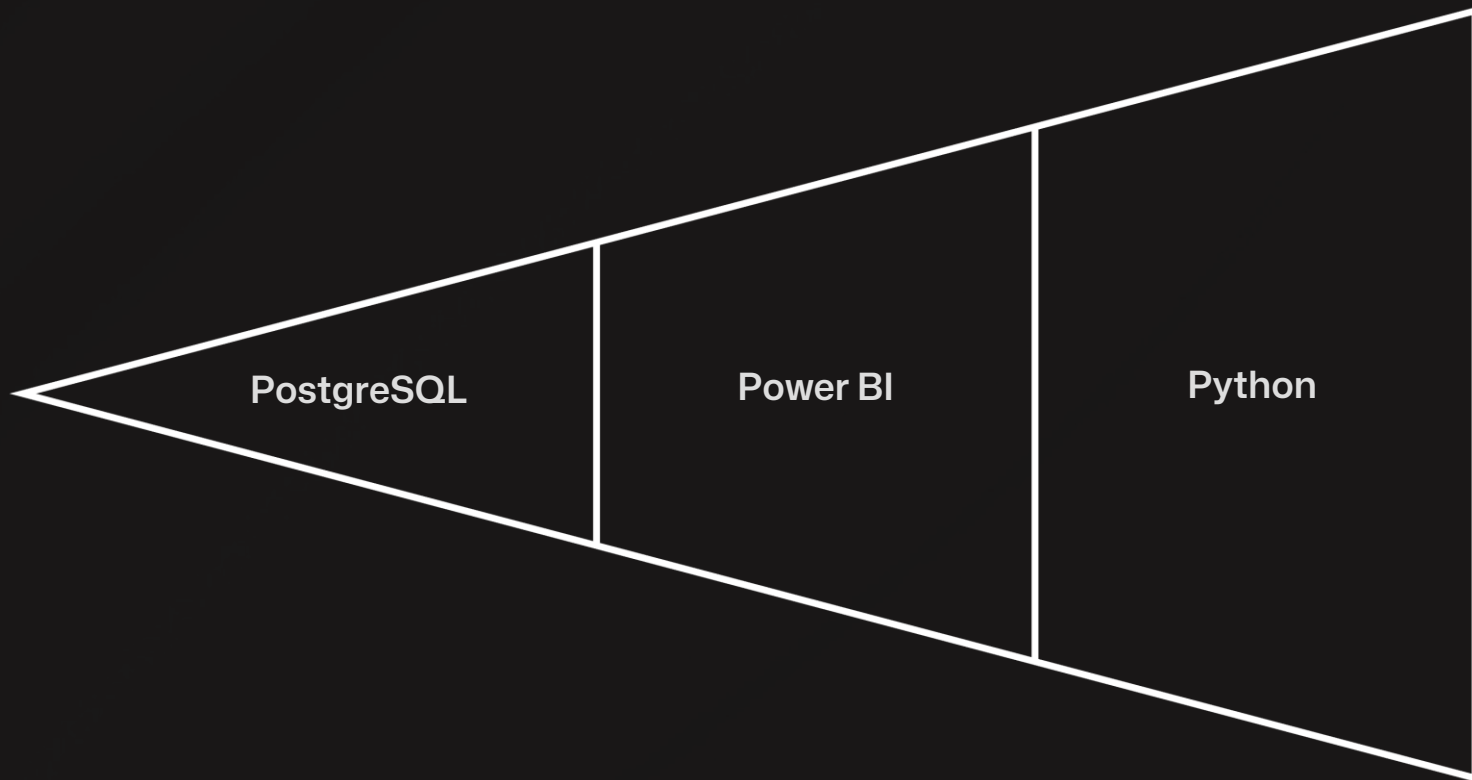
Where Is Spend Going?

Where is budget concentrated, and is the trend moving up or down?



Tech Stack & Analytics Pipeline

Four integrated layers transform raw shipment data into actionable business intelligence — from ingestion to AI-powered ad hoc queries.

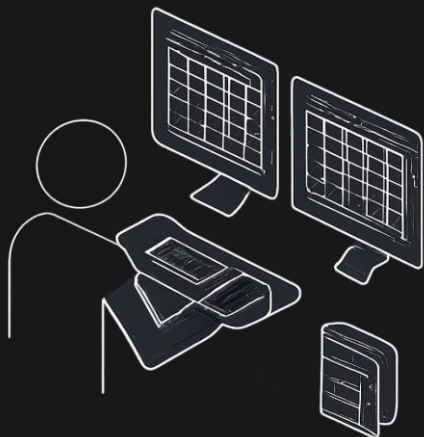


Each layer builds on the last — PostgreSQL ensures data quality, Python surfaces patterns, Power BI delivers self-service reporting, and the AI assistant handles follow-up questions that fall outside pre-built views.

Data Cleaning: Getting to Trustworthy Numbers

What We Fixed Before Analysis

- Removed **25 exact duplicate** order submissions
- Standardized casing and whitespace (" VIETNAM " → "Vietnam")
- Converted placeholder text in numeric columns to proper **NULL** values
- Nulled impossible values: zero unit price, negative quantities
- Cast all date fields to **DATE** type for lead-time calculations

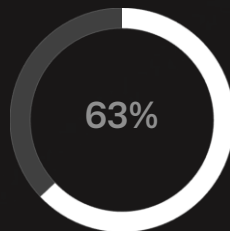


11 KPIs Analyzed

1	On-Time Delivery Rate Are shipments arriving by scheduled date?
2	Avg Lead Time (Planned vs Actual) How accurate is delivery forecasting?
3	Freight Cost % of Shipment Value Which modes are cost-efficient?
4	Freight Cost per Kilogram Logistics efficiency, normalized for size
5	Vendor Scorecard Spend, on-time rate, price volatility
6	Vendor Cost vs Reliability Are cheapest vendors least reliable?
7	Spend by Product Group / Country Where is budget concentrated?
8	Monthly Spend & Volume Trend Is demand and cost trending up or down?
9	Late Shipment Root Cause Which country + mode + vendor combos drive delays?
10	Quantity vs Unit Price Does bulk ordering earn a better price?

Delivery Performance: The Biggest Opportunity

Overall On-Time Delivery Rate

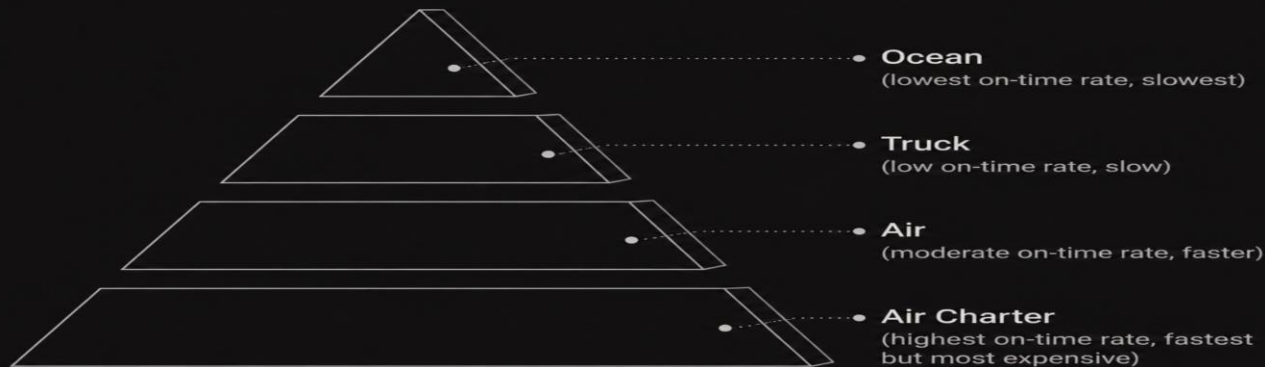


On-Time Rate

Only 63% of shipments arrive by their scheduled date — the single largest operational gap in the dataset.

Roughly 1 in 3 shipments misses its deadline. This is the primary area for logistics improvement and vendor accountability.

▲ Zimbabwe and Côte d'Ivoire have the lowest on-time rates and warrant an immediate root-cause review with the logistics team.



Shipment Mode Cost & Reliability Trade-Offs

Ocean Freight

Lowest cost per kg and lowest freight cost as % of shipment value — but the slowest and least reliable mode.

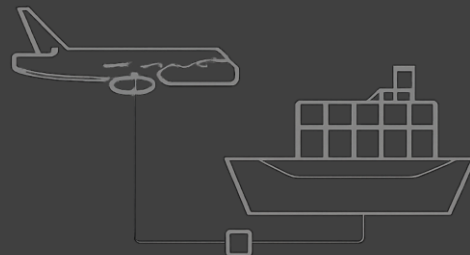
Air Charter

Costs roughly **2× more per kg** than ocean freight. Fastest and most reliable — ideal for time-critical health commodities.

Truck Freight

Moderate cost, but among the slowest and least reliable. Performance varies significantly by country route.

Key insight: The cost/speed trade-off is clear and quantifiable. Planners should evaluate route-by-route whether speed justifies the premium.



Vendor & Spend Insights

What the Data Reveals

Spend Is Well-Distributed

No single vendor dominates — top 5 vendors share spend fairly evenly, giving procurement negotiating leverage.

No Bulk Discount Pattern

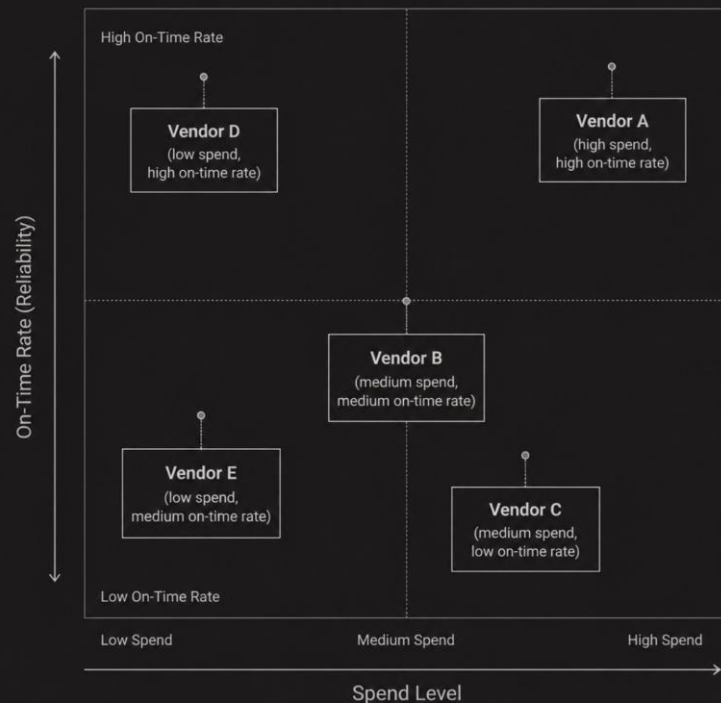
Quantity vs unit price correlation: 0.009. Average unit price stays flat at ~\$22-\$23 regardless of order size.

On-Time Rates Vary Meaningfully

Reliability differs between vendors at similar price points — a clear signal for contract renewals.

Delays Are Targeted, Not Systemic

Late shipments concentrate in specific country/mode/vendor combos (e.g., Guyana via Air) — a fixable pattern, not a general failure.



Vendor performance comparison across 5 top vendors: Vendor A (high spend, high on-time rate), Vendor B (medium spend, medium on-time rate), Vendor C (medium spend, low on-time rate), Vendor D (low spend, high on-time rate), Vendor E (low spend, medium on-time rate) — showing spend vs reliability trade-off

AI-Powered Data Assistant

A text-to-SQL interface that lets business users ask plain-English questions about shipment data — no SQL required. Built for ad hoc follow-ups that pre-built dashboards don't cover.



Pull Live Schema

Fetches real table structure from PostgreSQL



Generate SQL via Gemini

Question + schema sent to Gemini for a single SELECT statement



Validate Before Execute

Only SELECT allowed — DROP, DELETE, INSERT, ALTER are blocked unconditionally



Execute & Summarize

Query runs on Postgres; Gemini returns a plain-English summary



Example questions: "Which vendor in Vietnam has the worst on-time rate?" · "What's total freight cost for Ocean shipments over \$10k?"



Key Takeaways & Next Steps

What Leadership Should Act On

Fix the 37% late-delivery gap

Target Zimbabwe, Côte d'Ivoire, and specific vendor/mode combos — not a system-wide overhaul.

Revisit vendor pricing contracts

Bulk ordering yields no discount today. If volume pricing is intended, it isn't working.

Use mode trade-offs deliberately

Ocean is cheapest but slowest; Air Charter is 2× costlier but most reliable. Match mode to commodity urgency.

Deploy the AI assistant for ad hoc queries

Reduces analyst bottlenecks for follow-up questions outside pre-built KPI views.

